

Project Proposal: VoiceArmor-Edge

Real-Time On-Device Voice Spoof & Synthetic Call Detection
for Snapdragon-Powered HP PCs

Challenge: Snapdragon AI Lab Build & Present Challenge 2026

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Discipline: Electronics & Communication Engineering (ECE)

Target Hardware: Snapdragon X HP OmniBook PCs (Hexagon NPU) + Arduino UNO Q

1. Project Overview

AI tools can now clone a person's voice using just a short 3-second audio recording. Scammers are using these cloned voices to impersonate police officers, bank officials or family members to steal money -- a major issue right now with "Digital Arrest" scams in India.

Current security solutions rely on cloud servers to analyze audio. This does not work for live calls:

- Sending call audio over the internet is too slow to warn someone in real time.
- It requires an internet connection and leaks private conversations.
- It drains laptop battery and can make video calls laggy.

VoiceArmor-Edge solves this problem by running entirely on the laptop using the Snapdragon Hexagon NPU. It listens to incoming call audio in short 250ms chunks, converts the sound into frequency pictures (spectrograms) and runs a lightweight AI model to check if the voice is real or fake in under 15ms.

To give users a physical warning they cannot miss, VoiceArmor-Edge also connects via USB to an Arduino UNO Q board. If a fake voice is detected, the Arduino immediately turns on a bright red warning light and buzzer.

2. Real-World Need & Use Case

The Problem

- Voice Cloning Fraud: Fraudsters call victims pretending to be law enforcement or family members in distress, demanding urgent bank transfers.
- No Protection on Meeting Apps: Popular apps like Microsoft Teams, Zoom and WhatsApp Desktop have no built-in way to verify whether a voice is human or AI-generated.

Why It Must Run on the Snapdragon NPU

- Fast Response: Checking speech on the NPU takes less than 15ms, warning the user while the caller is still speaking.
- Complete Privacy: Audio never leaves the laptop. Everything works 100% offline, even in Airplane Mode.

- No Video Call Lag: The NPU handles the AI calculations, leaving the CPU and GPU free for the video call.
 - Low Battery Drain: The NPU uses very little power, allowing the protection to run in the background all day.
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3. Technical Implementation

A. Audio Feature Extraction (DSP)

When AI tools generate fake speech, they leave subtle irregularities in higher frequencies that natural human vocal cords do not produce.

- The application captures audio from the computer's audio output in real time.
- It slices the sound into 250ms chunks and generates a Log-Mel Spectrogram (a visual representation of sound frequencies over time).
- This step runs on the CPU and hands the data straight to the NPU.

B. Qualcomm AI Hub Model

- The classifier is adapted from standard audio models in the Qualcomm AI Hub catalog.
- INT8 Quantization: Model weights are converted from 32-bit to 8-bit integers. This speeds up processing on the Hexagon NPU and uses very little memory.
- The model is compiled using Qualcomm's official qai-hub toolchain to run as an ONNX model with QNN acceleration.

C. Running on Windows on Snapdragon

- The app runs as a native Windows service using onnxruntime with the onnxruntime-qnn execution provider.
- The model links directly with Qualcomm's QnnHtp.dll driver, routing computations to the Hexagon NPU cores.

D. Hardware Alert with Arduino UNO Q

In line with the Snapdragon AI Lab setup:

- The Snapdragon HP PC does the heavy AI computation.
 - The Arduino UNO Q connects via USB to provide a physical out-of-band indicator.
 - If a fake voice is detected, the PC signals the Arduino to turn on a red warning LED and buzzer.
 - Even if the user has their screen covered or minimized, the physical light immediately alerts them to hang up.
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4. Why This Approach Stands Out

- Practical & Focused: Instead of trying to do too many things, it solves one critical problem very well -- live synthetic voice detection.
- True Edge Computing: Proves why an on-device NPU is necessary. Cloud APIs cannot match the

sub-15ms speed and privacy required here.

- Physical Hardware Safety: Combining the HP PC with the Arduino UNO Q bridges software AI with real-world physical hardware alerts.
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5. Deployment on Snapdragon HP PCs

- Target Device: HP OmniBook laptops powered by Snapdragon X Elite / X Plus.
 - Operating System: Windows 11 on ARM64.
 - Lightweight: Uses less than 120 MB of RAM and under 3% CPU.
 - Works Offline: Operates completely in Airplane Mode without internet.
 - Easy to Use: Runs quietly in the Windows system tray and monitors call audio automatically.
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6. Project Presentation & Code

- Slide Deck: A clean 10-slide presentation explaining the problem, architecture, NPU integration and future roadmap.
 - Source Code: Full project code on GitHub, including the audio processing pipeline, NPU inference script and Arduino sketch.
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7. Conclusion

VoiceArmor-Edge is a focused, realistic and practical project that shows the real value of the Snapdragon Hexagon NPU. By combining digital signal processing, Qualcomm AI Hub model optimization and a physical hardware alert, it protects everyday users from growing AI voice scams safely and privately on their own laptops.